

The RWA Tokenization Stack

Technology, Governance, and Institutional Constraints

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November 2023

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Abstract

Tokenization of real-world assets (RWAs) is frequently presented as a technological stack composed of blockchains, smart contracts, and digital tokens. In practice, however, most tokenization initiatives fail to progress beyond pilots or limited deployments. This paper argues that these failures stem from a mischaracterization of tokenization as primarily a software problem. Drawing on institutional finance and market design perspectives, the paper proposes an alternative conception of the “RWA tokenization stack” as a multi-layered system in which technology is only one component. Legal enforceability, governance authority, fiduciary structures, and regulatory compatibility form higher-order layers that constrain and shape the viability of tokenized markets. By explicitly mapping these layers and their interactions, the paper provides a framework for understanding why many RWA tokenization efforts stall and what conditions are necessary for institutional adoption.

Keywords: real-world assets (RWA); asset tokenization; institutional market design; financial infrastructure; governance frameworks; legal enforceability; fiduciary accountability; regulatory compatibility; smart contracts; distributed ledger technology.

How to cite this paper: Sing, C. H. 2023. The RWA tokenization stack: Technology, governance, and institutional constraints. Working paper.

1. Introduction

Tokenization has emerged as a recurring theme in discussions of financial innovation, promising efficiency gains, broader access, and improved transparency across asset classes. Advances in distributed ledger technology and smart contracts have made it technically feasible to represent claims on real-world assets in digital form. Yet despite sustained experimentation across finance, infrastructure, and capital markets, institutional adoption of tokenized real-world assets (RWAs) remains limited.

Prevailing explanations for this gap often emphasize technical scalability, regulatory uncertainty, or insufficient market demand. While these factors are relevant, they obscure a more fundamental issue: tokenization is rarely constrained by technology alone. Institutions evaluate assets not merely by how ownership is recorded or transferred, but by how rights are enforced, decisions are governed, risks are allocated, and accountability is maintained over time. In this context, technical feasibility is a necessary but insufficient condition for adoption.

Many tokenization initiatives implicitly assume that institutional barriers will dissolve once software infrastructure matures. This assumption reflects a misdiagnosis of the problem. For most RWAs, the binding constraints are institutional rather than computational. Asset ownership and cash flows are constituted through legal agreements, governance arrangements, fiduciary responsibilities, and regulatory recognition that predate and supersede any technical implementation. Tokenization does not replace these structures; it operates within them.

This paper advances a different framing. It argues that successful RWA tokenization requires understanding that tokenization is not just a single technological layer, but instead, a stack of interdependent institutional layers where only one of which is technical. Failure to design and align the higher-order layers—legal enforceability, governance authority, fiduciary accountability, and regulatory compatibility—leads to systems that may function operationally yet remain unusable for institutional capital.

By reconceptualizing tokenization as an institutional stack rather than a software stack, the paper provides a framework for diagnosing why many tokenization efforts stall and for identifying the conditions under which tokenized RWAs can achieve durable, institutional adoption.

2. From “Technology Stack” to “Institutional Stack”

In software engineering, a “stack” typically refers to a layered arrangement of technical components—hardware, operating systems, middleware, applications, and interfaces—each abstracting complexity from the layer below. Applied uncritically to RWAs, this metaphor encourages an overly narrow focus on blockchains, protocols, and code execution, implicitly treating institutional features as external constraints to be addressed later.

For real-world assets, however, this framing is incomplete. Unlike purely digital assets, RWAs derive their economic meaning from off-chain institutions. Ownership, control rights, cash-flow

entitlements, and risk allocation are not created by code; they are constituted through legal contracts, governance mechanisms, and regulatory recognition. Tokenization does not originate these relationships—it reflects them.

As a result, a technology-centric view of the tokenization stack systematically underestimates the importance of non-technical layers. Many projects succeed at issuing tokens but fail to answer basic institutional questions: What exactly does the token represent? Who has authority to intervene when conditions change? How are disputes resolved? Who bears fiduciary responsibility if outcomes diverge from expectations? Absent clear answers, tokenized assets remain incompatible with institutional portfolios regardless of technical sophistication.

Accordingly, the relevant analytical question is not whether a token can be created, but whether the full stack supporting that token satisfies institutional requirements. This stack includes technical components, but it also encompasses governance authority, legal anchoring, fiduciary accountability, and regulatory integration. These higher-order layers determine whether tokenized claims are enforceable, investable, and scalable within existing market systems.

Reframing tokenization as an institutional stack therefore shifts attention away from isolated technical innovation toward system-level design. It highlights why progress at lower layers cannot compensate for weakness at higher layers, and why institutional adoption proceeds incrementally through structures that preserve accountability and control. This perspective sets the foundation for the layered framework developed in the next section.

3. The RWA Tokenization Stack: Six Layers

This paper proposes a six-layer framework for analyzing real-world asset tokenization. The layers are ordered from lowest to highest, not by technical sophistication, but by institutional significance. Lower layers enable functionality; higher layers determine investability. While progress at lower layers is often visible and rapid, institutional adoption depends on the alignment and robustness of the upper layers.

Crucially, the layers are not independent. Weakness or ambiguity at higher layers cannot be compensated for by improvements at lower ones. Tokenization efforts frequently fail not because technology malfunctions, but because institutional layers remain undefined, misaligned, or incompatible with existing fiduciary and regulatory systems.

Figure 1. The RWA Tokenization Stack

Layer 6: Regulatory and market integration
Layer 5: Fiduciary and risk accountability
Layer 4: Governance and decision authority
Layer 3: Asset representation and legal anchoring
Layer 2: Smart contracts and execution logic
Layer 1: Distributed ledger infrastructure

Layer 1: Distributed Ledger Infrastructure

At the base of the stack lies distributed ledger infrastructure, including public and permissioned blockchains. This layer provides cryptographic security, transaction ordering, data integrity, and resistance to unilateral manipulation. It establishes a shared record of transactions and ownership changes.

From an institutional perspective, however, this layer is largely commoditized. Competing ledger designs differ in throughput, governance, and openness, but the inability of blockchains to process transactions reliably is rarely the reason RWA initiatives stall. Ledger choice may influence operational characteristics, but it rarely determines whether an asset is investable.

This asymmetry explains why institutional attention quickly shifts away from infrastructure once baseline reliability is achieved. Distributed ledgers are necessary, but they do not resolve questions of ownership, authority, or responsibility.

Layer 2: Smart Contracts and Execution Logic

Smart contracts sit above the ledger, encoding rules for issuance, transfer, settlement, and basic lifecycle events. They automate standardized processes and reduce reliance on manual reconciliation or trusted intermediaries for routine actions.

However, execution logic is inherently constrained by what can be specified *ex ante*. Smart contracts excel in deterministic environments but perform poorly where discretion, judgment, and exception handling are required. Real-world assets routinely encounter events—regulatory changes, operational disruptions, force majeure, renegotiations—that cannot be fully anticipated or resolved through code alone.

When tokenization initiatives attempt to substitute automation for institutional discretion, they often introduce rigidity rather than resilience. As a result, smart contracts function best as procedural tools within a broader governance framework, not as replacements for it.

Layer 3: Asset Representation and Legal Anchoring

This layer defines what a token legally represents. Is it a direct ownership interest, a contractual claim, a beneficial interest held through a trust, or merely a reference to an off-chain agreement? These distinctions are not semantic; they determine enforceability, priority in insolvency, and the rights of token holders under stress.

Legal anchoring establishes the connection between on-chain representation and off-chain reality. It governs how tokenized claims are recognized by courts, regulators, custodians, and counterparties. Without clear legal documentation and jurisdictional recognition, tokens remain informational artifacts rather than enforceable financial instruments.

Many tokenization pilots stall at this layer. Tokens may trade or settle successfully on-chain, yet institutional investors remain unable to assess legal risk, creditor standing, or recovery rights. In such cases, technical success masks institutional failure.

Layer 4: Governance and Decision Authority

Governance determines who has authority to intervene when conditions deviate from expectations. This includes the power to pause transfers, modify parameters, unwind transactions, resolve disputes, or respond to external shocks.

In institutional markets, governance authority is deliberately residual and centralized. Boards, trustees, general partners, and regulators retain discretionary control precisely because not all contingencies can be specified in advance. Tokenization efforts that attempt to eliminate or obscure these authority structures undermine institutional trust.

Failures at this layer are particularly damaging. When authority is unclear, responsibility becomes diffuse, and institutions face uncertainty about who can act, when, and under what mandate. No amount of technical robustness at lower layers can compensate for ambiguous governance at higher ones.

Layer 5: Fiduciary and Risk Accountability

Institutions operate under fiduciary obligations that govern asset selection, custody, risk management, and oversight. This layer assigns responsibility for losses, ensures alignment of incentives, and enables auditability and escalation.

A tokenized structure without clearly defined fiduciary accountability may function operationally, yet remain unsuitable for institutional portfolios. If losses occur, institutions must be able to identify who is accountable, what duties were owed, and how breaches are remedied. Structures that distribute exposure without assigning responsibility violate these requirements.

Importantly, fiduciary accountability is not optional or negotiable. It is embedded in law, regulation, and professional standards. Tokenization designs that treat fiduciary considerations as secondary constraints rather than core design inputs are unlikely to scale.

Layer 6: Regulatory and Market Integration

At the top of the stack lies integration with regulatory frameworks and existing market infrastructure. Institutions must incorporate tokenized assets into compliance systems, reporting regimes, capital and liquidity calculations, and operational workflows.

This layer determines whether tokenized assets can move beyond isolated pilots. Regulatory compatibility is not a downstream consideration to be addressed after deployment; it is a precondition for scale. Tokenized assets that cannot be supervised, reported, or reconciled within existing systems remain marginal regardless of their technical merits.

Importantly, regulatory integration reflects institutional logic rather than regulatory conservatism. Supervisory frameworks evolve around identifiable entities, accountable managers, and enforceable rules. Tokenization succeeds where it preserves these anchors, not where it bypasses them.

Layer Interdependence and Institutional Dominance

The six layers together form an integrated system. Lower layers enable functionality, but higher layers determine legitimacy, trust, and scale. Misalignment across layers—such as advanced automation paired with weak legal anchoring or ambiguous governance—produces systems that are technically impressive yet institutionally unusable.

Understanding tokenization through this layered framework clarifies why institutional adoption proceeds unevenly and why progress at the technological frontier often fails to translate into deployable financial infrastructure. Tokenization is not constrained by what can be built, but by what can be governed.

4. Why Misalignment Causes Tokenization Failures

Most stalled RWA tokenization initiatives do not fail because of technological malfunction. Instead, they fail because progress across the tokenization stack is uneven. Lower layers—distributed ledgers and smart contracts—advance rapidly, while higher-order institutional layers remain weak, ambiguous, or unresolved. This vertical misalignment produces systems that operate technically but cannot be adopted at scale by institutional investors.

A common failure pattern involves sophisticated on-chain execution paired with insufficient legal anchoring. Tokens may be issued, transferred, and settled seamlessly, yet the underlying claims remain unclear in insolvency, unenforceable across jurisdictions, or subordinate to off-chain agreements. In such cases, technical clarity masks legal uncertainty, preventing institutions from assessing downside risk.

A second failure mode arises when automation outpaces governance design. Smart contracts encode deterministic rules, but real-world assets routinely encounter non-deterministic events—regulatory intervention, operational disruption, or contractual renegotiation. When authority to intervene is unclear or fragmented, institutions face unacceptable uncertainty regarding who can act, under what mandate, and with what liability. Automation without governance increases rigidity rather than resilience.

A third misalignment occurs when tokenization introduces tradability without corresponding fiduciary accountability. Tokenized claims may circulate widely, but responsibility for oversight, loss management, and compliance remains unclear. From an institutional perspective, liquidity without accountability is not an advantage; it is a risk amplifier.

These misalignments explain why many tokenization efforts remain trapped at the pilot stage. Institutions do not reject tokenization because it is novel, but because it redistributes risk without providing equivalent mechanisms for control and remediation. Until higher layers of the stack are aligned with lower ones, tokenization increases uncertainty rather than reducing it.

5. Implications for Institutional Adoption

Viewing tokenization through an institutional stack lens clarifies several recurring patterns in adoption behavior. These patterns are often interpreted as transitional frictions or temporary conservatism. In practice, they reflect stable institutional patterns shaped by fiduciary duty, governance requirements, and regulatory integration.

First, financial vehicles with pre-existing legal and governance structures adopt tokenization earlier than bespoke assets. Funds, trusts, and pooled vehicles already embed authority, accountability, and regulatory recognition. Tokenization in these contexts modifies record-keeping and settlement mechanics without altering the underlying institutional architecture. As a result, adoption can proceed incrementally rather than disruptively.

Second, permissioned or institution-led systems dominate early deployments. Institutions prioritize environments where authority, access, and accountability are clearly defined. Open or fully decentralized systems may offer technical advantages, but they often lack the governance clarity required for fiduciary participation. Control in this context is not a rejection of innovation, but a prerequisite for responsibility.

Third, traditional intermediaries persist rather than disappear. Custodians, administrators, trustees, and regulators remain essential because they occupy higher layers of the stack that technology alone cannot replace. Tokenization reconfigures these roles, but does not eliminate the functions they perform.

Taken together, these patterns suggest that institutional adoption of tokenized RWAs is not a linear progression toward decentralization. It is a selective integration process in which innovation is absorbed where it reinforces existing accountability structures and resisted where it undermines them.

6. Design Principles for Institutional Tokenization

Taken together, the preceding analysis suggests that successful real-world asset tokenization is not primarily a software design challenge. It is an institutional design problem in which technology must conform to existing structures of authority, accountability, and oversight. The following principles summarize the conditions under which tokenization can be absorbed into institutional portfolios rather than remaining confined to pilots or experimental deployments.

First, governance must precede automation. Tokenization should encode clearly defined authority structures rather than attempt to eliminate them. Decision rights, escalation mechanisms, and intervention powers must be specified *ex ante* and remain legible to fiduciaries and regulators. Automation enhances efficiency only when it operates within an explicit governance framework.

Second, legal enforceability must anchor technical representation. Tokens must correspond to claims that are enforceable under applicable legal regimes. This requires clear documentation of what a token represents, how rights are exercised, and how conflicts are resolved. Without firm legal anchoring, on-chain functionality does not translate into off-chain certainty.

Third, fiduciary accountability cannot be abstracted away. Institutional adoption requires that responsibility for outcomes be clearly assigned. Tokenization structures that distribute exposure without assigning accountability undermine the fiduciary obligations of institutions. Successful designs preserve clear lines of responsibility even as operational processes evolve.

Fourth, regulatory integration is a precondition for scale. Tokenized assets must be capable of fitting within supervisory, reporting, and compliance systems. Regulatory alignment is not a downstream optimization; it determines whether tokenized instruments can move beyond isolated use cases into systemically relevant markets.

These principles do not constrain innovation. Rather, they delineate the boundary between tokenization that enhances institutional markets and tokenization that merely shifts complexity without resolving underlying risks.

7. Conclusion

The limited institutional adoption of tokenized real-world assets is often attributed to technological immaturity or regulatory delay. This paper argues that such explanations misidentify the binding constraints. Tokenization initiatives stall not because they cannot be built, but because they are not designed to operate within the institutional systems that govern ownership, risk, and accountability.

By reframing tokenization as an institutional stack rather than a technology stack, the paper clarifies why progress at lower layers fails to compensate for weakness at higher ones. Distributed ledgers and smart contracts enable functionality, but legal anchoring, governance authority, fiduciary accountability, and regulatory integration determine whether tokenized assets are investable at scale.

This perspective also explains observed patterns of adoption. Tokenized funds and pooled vehicles advance more rapidly than bespoke assets because they preserve institutional continuity. Permissioned environments dominate early deployments because they align authority with responsibility. Intermediaries persist not due to inertia, but because they occupy roles that technology alone cannot replicate.

The implication is not that tokenization will remain marginal, but that its trajectory will be shaped by institutional logic rather than technological possibility. Durable adoption will occur where tokenization reinforces existing governance structures and stalls where it attempts to bypass them. For institutions, regulators, and market designers, the central challenge is therefore not how quickly assets can be tokenized, but under what conditions tokenization can be governed.

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